

Packet Tracer – VLAN Trunking and VTP Configuration

Objectives

Part 1: Build a Multi-Switch VLAN Topology

Part 2: Configure VLANs on the VTP Server

Part 3: Configure VTP Server and Client Modes

Part 4: Configure Trunk Links Between Switches

Part 5: Verify VLAN Propagation Through VTP

Part 6: Test Connectivity Within VLANs

Background / Scenario

In modern enterprise networks, VLANs are used to separate departments into different broadcast domains while still using the same physical switching infrastructure.

Managing VLANs individually on every switch can become time-consuming. Cisco's **VLAN Trunking Protocol (VTP)** helps automate VLAN distribution by allowing one switch to act as a VTP Server while other switches receive VLAN information as VTP Clients.

In this lab, you will configure VLANs, establish trunk links, configure VTP, and verify that VLAN information is automatically synchronized between switches.

This lab focuses on:

- VLAN creation
- Access port assignment
- Trunk configuration
- VTP Server and Client configuration
- VLAN propagation
- Switch management and verification

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 2 Switches (2960)
- 4 PCs

Rename them:

Switch0 → S1

Switch1 → S2

PC0 → PC-A

PC1 → PC-B

PC2 → PC-C

PC3 → PC-D

Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Connect:

PC-A → S1 Fa0/1

PC-B → S1 Fa0/2

PC-C → S2 Fa0/1

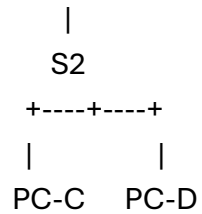
PC-D → S2 Fa0/2

S1 Fa0/24 → S2 Fa0/24

Topology:

PC-A PC-B

```
|           |
+----S1----+
      |
    Trunk Link
```



Wait for all links to turn green.

Part 2: Configure VLANs

Step 1: Configure VLANs on S1

Access S1 CLI.

enable

configure terminal

vlan 10

name SALES

vlan 20

name HR

end

Verify:

show vlan brief

Expected:

VLAN 10 SALES

VLAN 20 HR

Step 2: Assign Access Ports

Configure PC-A and PC-C for VLAN 10.

Configure PC-B and PC-D for VLAN 20.

On S1:

configure terminal

interface fa0/1

switchport mode access

switchport access vlan 10

interface fa0/2

switchport mode access

switchport access vlan 20

end

Part 3: Configure VTP

Step 1: Configure S1 as VTP Server

configure terminal

vtp domain COMPANY

vtp mode server

vtp password cisco

end

Verify:

show vtp status

Expected:

VTP Operating Mode: Server

Domain Name: COMPANY

Step 2: Configure S2 as VTP Client

configure terminal

vtp domain COMPANY

```
vtp mode client
```

```
vtp password cisco
```

```
end
```

Verify:

```
show vtp status
```

Expected:

VTP Operating Mode: Client

Domain Name: COMPANY

Part 4: Configure Trunking

Step 1: Configure Trunk on S1

```
configure terminal
```

```
interface fa0/24
```

```
switchport mode trunk
```

```
end
```

Step 2: Configure Trunk on S2

```
configure terminal
```

```
interface fa0/24
```

```
switchport mode trunk
```

```
end
```

Step 3: Verify Trunk

```
show interfaces trunk
```

Expected:

Port	Mode	Encapsulation	Status
------	------	---------------	--------

Fa0/24	trunk	802.1Q	trunking
--------	-------	--------	----------

Part 5: Verify VLAN Propagation

Step 1: Check VLAN Database on S2

On S2:

```
show vlan brief
```

Expected:

```
10 SALES
```

```
20 HR
```

Notice:

You never manually created VLANs on S2.

VTP automatically propagated them from S1.

Step 2: Assign Ports on S2

```
configure terminal
```

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
interface fa0/2
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
end
```

Part 6: Configure IP Addresses

Step 1: Configure VLAN 10 Hosts

PC-A

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

PC-C

IP Address: 192.168.10.20

Subnet Mask: 255.255.255.0

Step 2: Configure VLAN 20 Hosts

PC-B

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

PC-D

IP Address: 192.168.20.20

Subnet Mask: 255.255.255.0

Part 7: Test Connectivity

Step 1: Test VLAN 10 Communication

From PC-A:

ping 192.168.10.20

Expected:

Reply from 192.168.10.20

Success.

Step 2: Test VLAN 20 Communication

From PC-B:

ping 192.168.20.20

Expected:

Reply from 192.168.20.20

Success.

Step 3: Test Inter-VLAN Communication

From PC-A:

ping 192.168.20.10

Expected:

Request Timed Out

Why?

Because there is no router performing inter-VLAN routing.

Verification Commands

View VLANs

show vlan brief

View Trunks

show interfaces trunk

View VTP Status

show vtp status

View Running Configuration

show running-config

Understanding the Technologies

VLAN

Separates broadcast domains.

VLAN 10 = SALES

VLAN 20 = HR

Devices in different VLANs are logically separated.

Trunk

Carries multiple VLANs over one physical link.

Without trunking:

1 VLAN = 1 Cable

With trunking:

Many VLANs = 1 Cable

VTP

Automatically distributes VLAN information.

S1 (Server)

|

Trunk

|

S2 (Client)

Create VLAN once.

Receive everywhere.

Troubleshooting Tips

If VLANs do not appear on S2:

✓ Verify VTP domain names match

show vtp status

✓ Verify passwords match

✓ Verify trunk link is active

show interfaces trunk

✓ Verify VTP modes

show vtp status

✓ Verify VLAN exists on server

show vlan brief

What You Learned

- How VLANs separate networks
- How to configure access ports
- How to configure trunk links
- How VTP distributes VLAN information
- How VTP Server and Client operate
- How VLANs affect connectivity
- Why inter-VLAN routing requires a router or Layer 3 switch

Challenge Extensions (Optional)

Challenge 1: Add VLAN 30

On S1:

vlan 30

name IT

Verify:

show vlan brief

Check S2.

Did VLAN 30 automatically appear?

Challenge 2: Configure S2 as Transparent Mode

vtp mode transparent

Create VLAN 40 on S1.

Questions:

- Does S2 learn VLAN 40?
 - How does Transparent Mode behave?
-

Challenge 3: Restrict Allowed VLANs on the Trunk

On S1:

interface fa0/24

switchport trunk allowed vlan 10

Verify:

VLAN 20 traffic should fail

VLAN 10 traffic should succeed

Challenge 4: Configure Switch Management IP

On S1:

interface vlan 10

ip address 192.168.10.254 255.255.255.0

no shutdown

Verify:

ping 192.168.10.254

This allows remote management of the switch.

Challenge 5: Observe VLAN Tags in Simulation Mode

Switch Packet Tracer to:

Simulation Mode

Generate traffic between:

PC-A ↔ PC-C

Observe:

- VLAN tags on trunk links
- Frame forwarding decisions
- Broadcast containment inside VLANs